

15.439 Quantitative Investment Management

Assignment 4

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Setup and Notation

Let $\mu = 0.08$ (annualized expected return) and $\sigma = 0.04$ (annualized standard deviation), so the annualized Sharpe ratio is $SR = \mu/\sigma = 2.0$.

Because returns are assumed i.i.d. Normal, for any horizon of length T (measured in years):

$$R_T \sim \mathcal{N}(\mu T, \sigma^2 T), \quad \sigma_T = \sigma\sqrt{T}.$$

We use 252 trading days per year. Let Φ denote the standard Normal CDF and ϕ its density.

Part A — Loss Probabilities

The probability of a loss over horizon T is

$$\mathbb{P}(R_T < 0) = \Phi\left(\frac{-\mu T}{\sigma\sqrt{T}}\right) = \Phi\left(\frac{-\mu\sqrt{T}}{\sigma}\right) = \Phi(-SR\sqrt{T}).$$

(i) **Annual** ($T = 1$):

$$\mathbb{P}(\text{loss}) = \Phi(-2.0) = \boxed{0.0228}.$$

(ii) **Quarterly** ($T = 0.25$):

$$\Phi(-2.0\sqrt{0.25}) = \Phi(-1.0) = \boxed{0.1587}.$$

(iii) **Monthly** ($T = 1/12$):

$$\Phi(-2.0/\sqrt{12}) = \Phi(-0.5774) = \boxed{0.2819}.$$

(iv) **Daily** ($T = 1/252$):

$$\Phi(-2.0/\sqrt{252}) = \Phi(-0.1260) = \boxed{0.4499}.$$

(v) **\$500M quarterly PnL**. Taking expectations over the physical probability measure $\mathbb{P}$ in a classical filtered prob space $(\Omega, \mathcal{F}, \mathcal{F}_{t \in I}, \mathbb{P})$, we have, for $W = \$500\text{M}$:

$$\mathbb{E}[\text{PnL}_Q] = W \cdot \mu \cdot 0.25 = 500\text{M} \times 0.02 = \$10,000,000,$$

$$\sigma_{\text{PnL}} = W \cdot \sigma\sqrt{0.25} = 500\text{M} \times 0.02 = \$10,000,000.$$

Hence the $\pm 2\sigma$ band is

$$\boxed{\$10\text{M} \pm \$20\text{M}}, \quad \text{i.e. } [-\$10\text{M}, \$30\text{M}].$$

Even with a Sharpe of 2.0, the daily loss probability is $\approx 45\%$; the signal-to-noise ratio $\mu_d/\sigma_d = SR/\sqrt{252} \approx 0.126$ is tiny at high frequency.

Part B — Winning Rate, Up/Down Ratio, and Sharpe Ratio

Let $\mu_d = \mu/252$ and $\sigma_d = \sigma/\sqrt{252}$, so $R_d \sim \mathcal{N}(\mu_d, \sigma_d^2)$.

Winning rate.

$$w = \mathbb{P}(R_d > 0) = \Phi\left(\frac{\mu_d}{\sigma_d}\right) = \Phi\left(\frac{SR}{\sqrt{252}}\right).$$

For $SR = 2$: $w = 0.5501$.

Up/Down ratio. Using the truncated Normal expectations:

$$\mathbb{E}[R_d \mid R_d > 0] = \mu_d + \sigma_d \frac{\phi(-\mu_d/\sigma_d)}{\Phi(\mu_d/\sigma_d)},$$

$$\mathbb{E}[|R_d| \mid R_d < 0] = -\mu_d + \sigma_d \frac{\phi(-\mu_d/\sigma_d)}{\Phi(-\mu_d/\sigma_d)}.$$

The up/down ratio is $U/D = \mathbb{E}[R_d \mid R_d > 0] / \mathbb{E}[|R_d| \mid R_d < 0]$.

For the base case: $U/D = 1.1216$.

B.i — Varying σ from 2% to 12% (step 0.25%), $\mu = 8\%$ fixed

Selected rows (full table produced by simulation code):

σ	Win Rate	Up/Down	Sharpe
0.0200	0.5995	1.2577	4.00
0.0400	0.5501	1.1216	2.00
0.0600	0.5335	1.0795	1.33
0.0800	0.5251	1.0591	1.00
0.1000	0.5201	1.0470	0.80
0.1200	0.5167	1.0390	0.67

As σ increases (holding μ fixed), the Sharpe ratio, winning rate, and up/down ratio all decrease together.

B.ii — Varying μ from 2% to 12% (step 0.25%), $\sigma = 4\%$ fixed

Selected rows:

μ	Win Rate	Up/Down	Sharpe
0.0200	0.5126	1.0291	0.50
0.0400	0.5251	1.0591	1.00
0.0600	0.5376	1.0899	1.50
0.0800	0.5501	1.1216	2.00
0.1000	0.5626	1.1542	2.50
0.1200	0.5749	1.1877	3.00

As μ increases (holding σ fixed), all three quantities increase.

B.iii — Interpretation

Under the i.i.d. Normal assumption, the daily Sharpe ratio is $SR_d = \mu_d/\sigma_d = SR/\sqrt{252}$. The winning rate and up/down ratio are both *monotone increasing* functions of SR_d alone:

$$w = \Phi(SR_d), \quad U/D = \frac{SR_d + \phi(-SR_d)/\Phi(SR_d)}{-SR_d + \phi(-SR_d)/\Phi(-SR_d)}.$$

So all three metrics are effectively determined by a single parameter SR_d . Despite a very high annualized Sharpe of 2, the daily winning rate is barely above 55% and the up/down ratio is only 1.12.

Assumptions driving this: (1) returns are i.i.d. (no serial correlation, no regime changes), (2) returns are Normal (no skew or fat tails), (3) the Sharpe ratio is constant across time. In practice, strategies with option-like payoffs, momentum effects, or fat tails can produce very different winning rate / up-down ratio relationships for the same Sharpe.

Part C — Drawdown Simulations

A *drawdown* is the number of consecutive days during which cumulative PnL remains below its running maximum. There is no closed-form solution; we use Monte Carlo simulation with 100,000 paths.

Daily returns: $R_d \sim \mathcal{N}(\mu_d, \sigma_d^2)$ with constant dollar exposure (PnL is additive).

C.i — 5-year horizon, 90-day drawdown, $SR = 2$

$$\mathbb{P}(\text{90-day drawdown in 5 years}) \approx 0.8957.$$

Despite an excellent Sharpe ratio of 2.0, a 90-day drawdown is nearly certain over a 5-year period.

C.ii — 10-year horizon

$$\begin{aligned}\mathbb{P}(\text{90-day DD in 10y}) &\approx 0.9905, \\ \mathbb{P}(\text{120-day DD in 10y}) &\approx 0.9186.\end{aligned}$$

As the observation window lengthens, the probability of observing a drawdown of any fixed length converges to 1. Intuitively, positive drift does not eliminate drawdowns; it only makes recoveries more likely on average. Over a long enough horizon, however, the process still experiences sufficiently adverse stretches of returns to remain below its previous running maximum for any prescribed finite length.

Namely, letting $D_T^{(L)}$ denote the event that, by time T , the P&L process has experienced at least one drawdown of length at least L trading days, we claim that

$$\lim_{T \rightarrow \infty} \mathbb{P}(D_T^{(L)}) = 1 \quad \text{for every fixed } L \in \mathbb{N}^*.$$

In fact, one can strengthen this to the almost sure statement

$$\mathbb{P}\left(\bigcup_{T \geq 1} D_T^{(L)}\right) = 1.$$

The main idea is that, under the i.i.d. Gaussian assumption, there is strictly positive probability that any given block of L consecutive days is sufficiently poor that the cumulative P&L stays below its prior running maximum throughout the entire block; since disjoint blocks are independent and occur infinitely often, the probability that this never happens is zero.

Shortening the commitment period reduces the probability, but even over 5 years a 90-day drawdown is overwhelmingly likely.

C.iii — Drawdown Table (1-year lock-up, varying SR)

DD (days)	SR=−1.0	SR=−0.5	SR=0.0	SR=0.5	SR=1.0	SR=1.5	SR=2.0	SR=2.5	SR=3.0
1	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
3	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
5	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
10	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
20	1.000	1.000	1.000	1.000	0.999	0.997	0.993	0.986	0.967
40	0.999	0.997	0.991	0.974	0.949	0.897	0.821	0.723	0.601
60	0.990	0.977	0.944	0.888	0.804	0.691	0.558	0.423	0.294
90	0.952	0.902	0.815	0.700	0.559	0.417	0.287	0.177	0.098
120	0.883	0.792	0.664	0.519	0.374	0.242	0.146	0.075	0.035

Part D — Discussion**Drawdowns as a measure of skill**

The table above reveals that drawdowns are extremely poor discriminators of manager skill, especially over short evaluation horizons.

- 1. Even excellent managers experience long drawdowns.** A manager with $SR = 2$ (top-decile skill) has a 29% chance of a 90-day drawdown and a 15% chance of a 120-day drawdown within a single year.
- 2. Drawdowns barely distinguish skill from no skill.** At the 90-day row, the probability only falls from 0.82 ($SR = 0$) to 0.29 ($SR = 2$). A Bayesian investor starting with a diffuse prior over Sharpe ratios cannot sharply update based on a single drawdown observation.

3. **Short drawdowns are uninformative.** For drawdowns ≤ 20 days, probabilities are near 1.0 across all Sharpe ratios. These events carry essentially zero information about skill.

Implications for hedge fund managers

- **Lock-up periods are essential.** If investors redeem after a 90-day drawdown, even a $SR = 3$ fund faces $\sim 10\%$ chance of forced liquidation per year. Managers should seek the longest lock-ups possible.
- **Manage investor expectations.** Present the drawdown probability table to investors upfront; educate them that drawdowns are *expected*, not evidence of broken strategy.
- **Diversify across independent strategies.** By the Fundamental Law of Active Management, $IR = IC\sqrt{BR}$. Running multiple independent bets reduces portfolio-level drawdown length for the same aggregate Sharpe, because drawdowns in one strategy are unlikely to coincide perfectly with drawdowns in another.

How long before an investor can judge skill?

The t -statistic for the Sharpe ratio after n years is approximately $SR\sqrt{n}$. To achieve $t = 2$ with $SR = 1$, one needs $n = 4$ years. With $SR = 0.5$, one needs 16 years. This is the core problem: the time required to statistically distinguish skill from luck is on the order of years to decades.

Winning rates and up/down ratios face the same problem. A winning rate of 55% versus 53% requires thousands of observations to distinguish reliably, and both are consistent with meaningful differences in annualized Sharpe.

Connection to Limits of Arbitrage

Shleifer and Vishny (1997) argue that even if a manager has genuine skill (positive- α strategy), **short-term losses trigger investor withdrawals**, forcing liquidation precisely when expected returns are highest. Our drawdown analysis quantifies this: a skilled manager ($SR = 2$) still faces near-certain multi-month drawdowns, which—in the absence of lock-ups—would trigger capital flight and destruction of the arbitrage.

This creates a feedback loop: the possibility of forced liquidation makes the strategy riskier than its fundamental volatility suggests, which in turn makes investors more skittish, amplifying the problem.

Connection to the Fundamental Law of Active Management

The Fundamental Law ($IR = IC \times \sqrt{BR}$) implies that a manager with moderate skill (IC) but many independent bets (BR) can achieve a high information ratio. This directly maps to drawdown resilience: more independent bets mean diversification across time and assets, reducing the variance of cumulative PnL and thus shortening expected drawdown lengths.

Reporting frequency: pros and cons

Pros of high-frequency reporting. Transparency builds trust; investors can verify risk limits; early detection of style drift or operational problems.

Cons of high-frequency reporting. Daily or weekly returns have extremely low signal-to-noise ($SR_d \approx 0.13$ even for $SR = 2$), so investors observe near-coin-flip outcomes. This leads to false inference, myopic loss aversion (Benartzi and Thaler, 1995) and pressure on managers to reduce risk during drawdowns, harming long-run returns.

Optimal practice. Report risk metrics (exposure, leverage, VaR) frequently, but present performance statistics at lower frequencies (quarterly or annually) with appropriate context about expected drawdown distributions.

References

Shlomo Benartzi and Richard H. Thaler. Myopic Loss Aversion and the Equity Premium Puzzle. *Quarterly Journal of Economics*, 110(1):73–92, 1995.